HANNA PARK

US Citizen | hpark4@caltech.edu | linkedin.com/in/park-hanna

Mechanical engineer interested in fluid and continuum mechanics, mathematical modeling of complex fluids and soft matter, and robotic locomotion, with a background in aerodynamics, flight control, and robotic systems.

EDUCATION

Caltech | *B.S. in Mechanical Engineering* | ***GPA:** 4.2 / 4.3* | Expected Jun 2027 — Pasadena, CA

***Relevant Coursework:** Robotics, Dynamics & Control, Mechanical Prototyping, Design & Fabrication, CFD, Diff. Equations & Transform Methods*

***In Progress (2026–27):** Advanced Fluid Mechanics, Machine Learning, Linear Systems & Control, Data-Driven Modeling & Dimensional Analysis*

University of Cambridge, St John's College | *Engineering* | Oct - Dec 2025 — Cambridge, United Kingdom

***Completed Courses (graduate-level):** Aircraft Aerodynamics & Design; Stability & Control; Nuclear Reactor Engineering; Technology Management*

AWARDS

Henry Ford II Scholar Award | *Caltech Engineering and Applied Science, strongest GPA in Mechanical Engineering* — 2026

Jack E. Froehlich Memorial Award Nominee | *Top 5% GPA, with selection based on academics and research* — 2026

Korean Honor Scholarship (Highest Honors) | *Top undergraduate recipient; awarded by the Korean government* — 2026

SKILLS

Robotics & Flight Control: flight-control integration, ROS, MAVLink/MAVROS, ArduPilot/PX4, QGroundControl, Mission Planner
Simulation & Programming: OpenFOAM, ANSYS Mechanical, Simulink, Python, MATLAB, C/C++, Linux, Git
Experimental Tools & Testing: Instron, Mecmesin, cryogenic pressure testing, OptiTrack, wind-tunnel testing, PIV/dye visualization
Mechanical Design: SolidWorks, Siemens NX, engineering drawings, machining, 3D printing, soldering, wiring
Languages: English (native), Korean (native), Japanese (proficient), Spanish (conversational)

EXPERIENCE

SpaceX Propulsion Engineering Internship — June 2026 – September 2026
Falcon Fluids and Mechanical Systems Team — *Hawthorne, CA*
- Routed pipes and ran ANSYS random vibration/fatigue analysis using PSD data to calculate apparent frequency for reliability.
- Designed and led cryogenic pressurized and tensile testing for quick disconnects, including custom fixture development.

Senior Thesis – M4 Morphobot: Air-Ground Locomotion and Thrust-Deflector Design — January 2026 – Present
Gharib Group, California Institute of Technology | Co-mentored with NASA Jet Propulsion Laboratory — *Pasadena, CA*
- Building drone platform from initial design (CAD, fab/assembly, propulsion/avionics integration) for M4 air-ground testing.
- Designing thrust-biasing deflectors; using ANSYS and OpenFOAM CFD, along with bench testing, to quantify thrust and drag.

Gharib Group Intern – AI-Driven Flight Control Systems for Fixed-Wing UAVs in Turbulence — October 2024 – January 2026
Center for Autonomous Systems and Technologies, Graduate Aerospace Laboratories of Caltech (GALCIT) — *Pasadena, CA*
- Integrated embedded sensing, flight-control, actuator, and ground-station systems for fixed-wing aerial robot testing.
- Built ROS/MAVLink workflows for synchronized pressure, IMU, and autopilot data during controlled free-flight experiments.
- Designed and wind-tunnel calibrated five-hole Pitot probe hardware for real-time aerodynamic flow measurement.

Gharib Group Researcher – Low Reynolds Number Water Tunnel — February 2024 – June 2024
Graduate Aerospace Laboratories of Caltech (GALCIT), California Institute of Technology — *Pasadena, CA*
- Collaborated on water-tunnel hardware, turbine-array fixtures, electronics, and power configuration.
- Implemented Arduino/PWM turbine-array control and supported PIV/dye validation of shear-layer/vortex-shedding behavior.

Real-Time Vision & Motion Control System Project — September 2023 – December 2023
California Institute of Technology — *Pasadena, CA*
- Pair-programmed a real-time camera-guided tracking system using Python/OpenCV target detection and HEBI actuator control.

Summer Undergraduate Research Fellowship (SURF) on Kinetic Inductance Detectors — June 2024 – September 2024
Golwala Group, California Institute of Technology — *Pasadena, CA*
- Redesigned cryogenic detector hardware through CAD, circuit-geometry updates, mounting design, and wire-bonded assembly.
- Developed a Python resonance-fitting pipeline for superconducting resonator analysis, reducing fit time from 20 hr to 20 min.

Wellatex Inc. – Assembly Machine Management and Product Design Lead — January 2019 – Present
- Assembled industrial packing machinery and designed palletized shipping boxes for 50+ product dimensions.

TEACHING, LEADERSHIP, AND ACTIVITIES

Incoming Teaching Assistant, ME 50a & ME 50b (Prof. Michael Mello) | *Caltech* — January 2027 – June 2027
- CFD & Experimental Fluid Mechanics (50a); FEA & Experimental Solid Mechanics (50b).

Teaching Assistant, Biomechanics (Prof. Michael Dickinson) | *Caltech* — January 2025 – March 2025
- Prepared and graded exams; held office hours to support student understanding of solid/fluid mechanics and materials.

Caltech Air and Outer Space Club | *NASA BIG Idea Challenge Finalist* — September 2023 – February 2024
- Developed proposal for an inflatable lunar landing shield, modeling dust/regolith behavior during landing.

Caltech Admissions Ambassador Professional Development Lead — May 2024 – Present
- Conduct tours and serve as a student panelist; deliver audience-specific Q&A on curriculum, research, and student life.

Interests | *Cello, Caltech chamber music and orchestra, and baking*